SERVER:
COMPOSABLE MEMORY
SYSTEMS (CMS)



Cost-Efficient LLM Inference via Peer-to-Peer GPU-to-Memory Expansion Datapath

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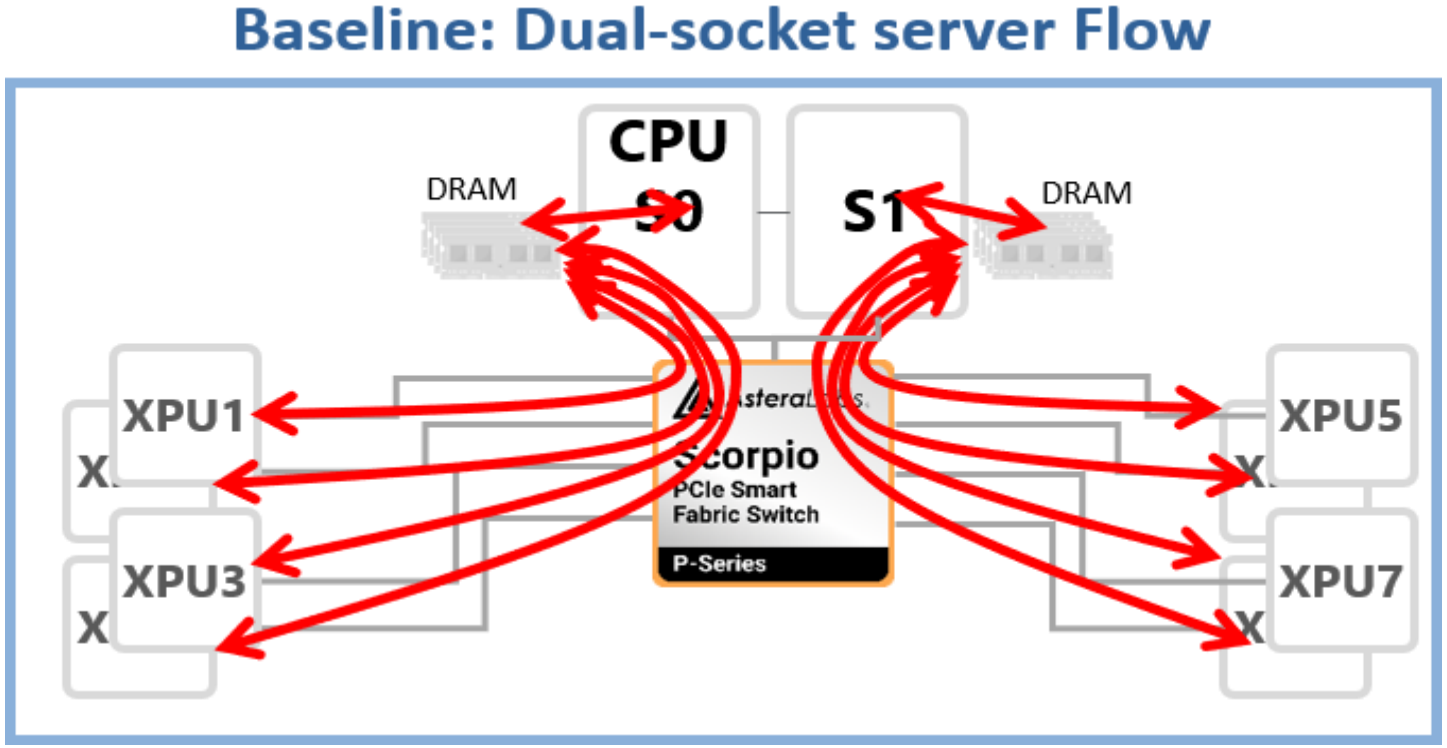
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OCP PLATINUM Member

Motivation

Inference can be accelerated by offloading the key-value cache (KV-CO) to external memory: conventionally, to the DRAM attached directly to the host CPU.

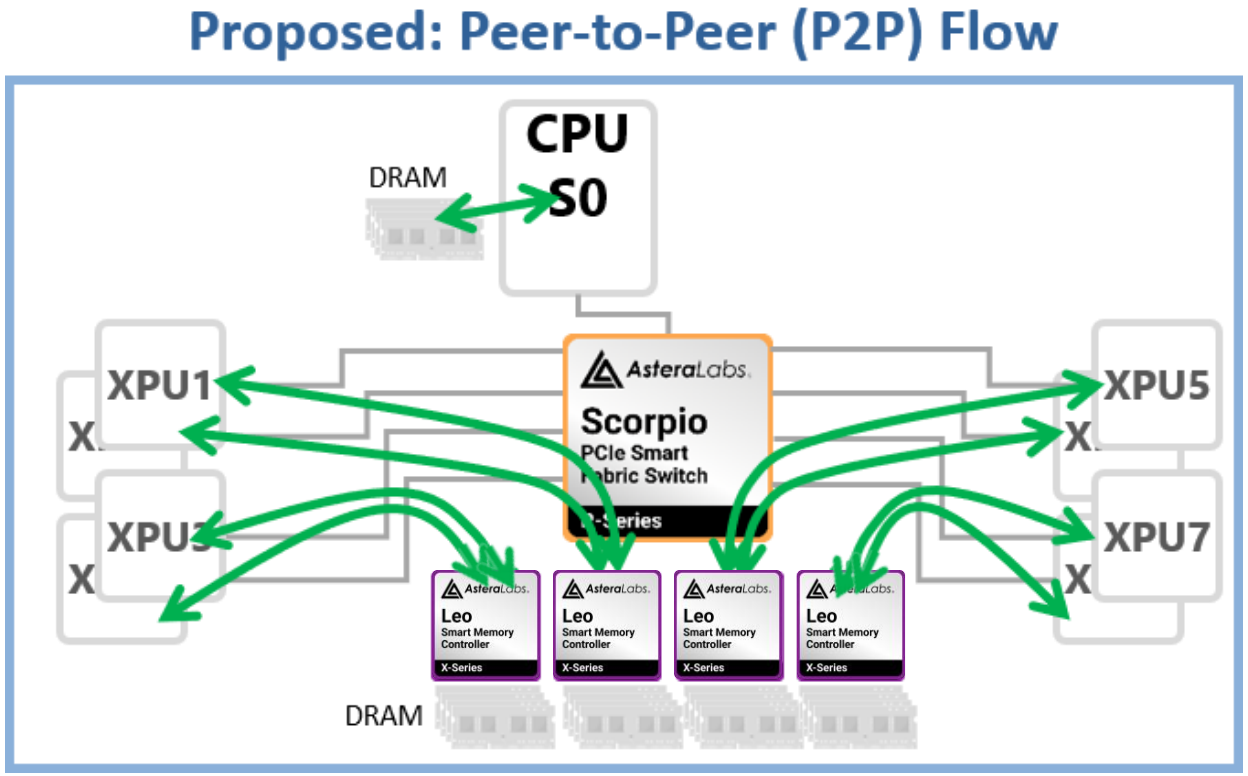
This typical setup limits the available memory capacity and bandwidth accessible by both the XPU AI accelerators and the CPU.



Proposed Fabric Peer-to-Peer Flow

We propose leveraging emerging fabrics to add memory controllers, routing the KV-CO XPU traffic directly to them in a peer-to-peer (P2P) fashion. This:

- Distributes traffic and avoids bottlenecks,
- Provides a clear path for scaling capacity and bandwidth,
- Allows dedicated CPU BW for memory-intensive Agentic workloads
- Potentially eliminates the need for a second CPU socket



Experimental Setup

- The proposed P2P setup that was deployed for evaluation, consists of:
  - 4 GPUs - NVIDIA RTX6000 Pro (96GB HBM)
  - 1 CPU socket - Intel Xeon Platinum 8468
  - 4 Memory controllers - Astera Labs Leo Smart Memory Controllers, each with 256GB of DDR5 5600MT/s
  - 1 PCIe switch - Astera Labs Scorpio (320 lanes) Smart Fabric Switch

Benchmark:	vLLM Multi-turn
Inference engine: vLLM 0.19 KV-CO Mgr:	LMCache 0.4.3*
Model:	Qwen3-14B
Baseline CPU DRAM	**P2P** Leo Memory

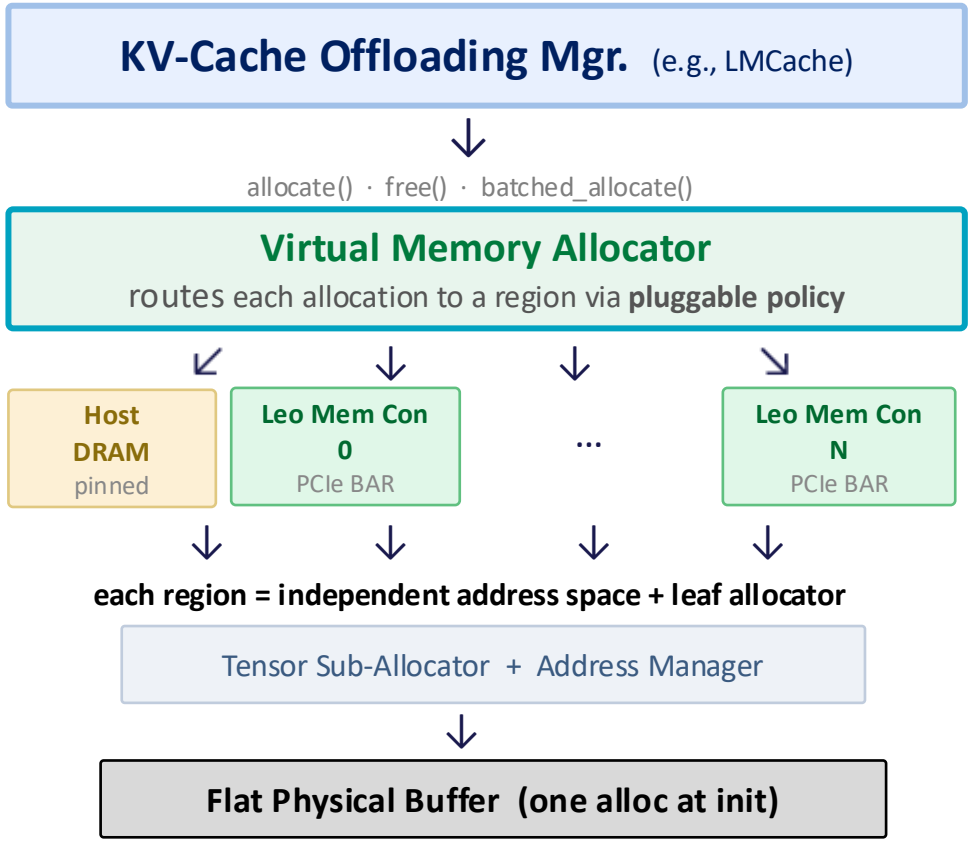
We evaluated vLLM KV-CO acceleration,
using 4 Qwen3 instances and the vLLM multi-turn benchmark.

Parameters:

• Initial input tokens range:	1000-2000 (short) 2000-3500 (med) 5000-6500 (med-long)
• Output tokens range:	100-150
• Turns:	3-8
Tokens per turn	120-180
• Variable batch size	10-60
• Copy mechanism, chunk size	DMA, 128 tokens

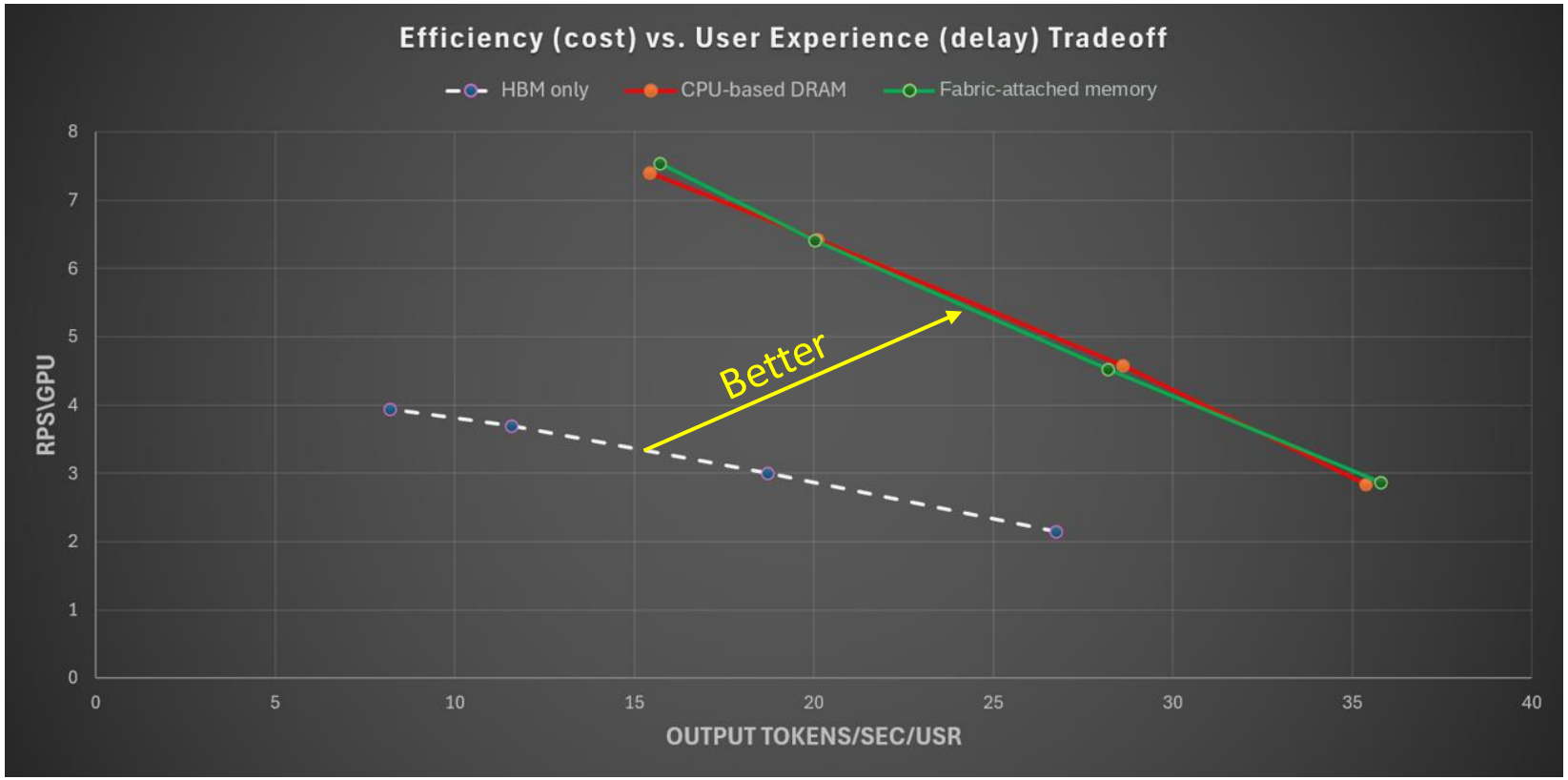
Enhancement to the KV-CO Manager

Multi-region virtual memory layer for heterogeneous KV-cache tiers: host DRAM + memory controllers (Astera Labs Leo) behind a single allocator interface, with multiple allocation policies.

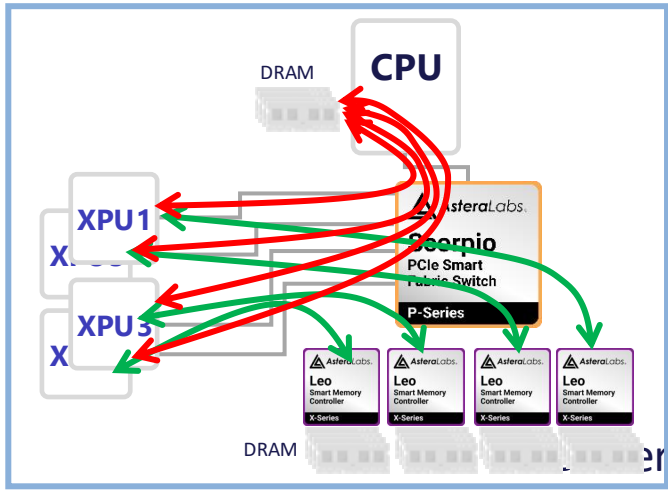


KV-CO Results – 1

Performance identical to local host memroy
for the same capacity and **short input prompts**



Legend:
Traditional KV-CO
Fabric-attached memory



Parameters:

- **Short input prompts** (1-2K)
- 128GB DRAM per vLLM instance

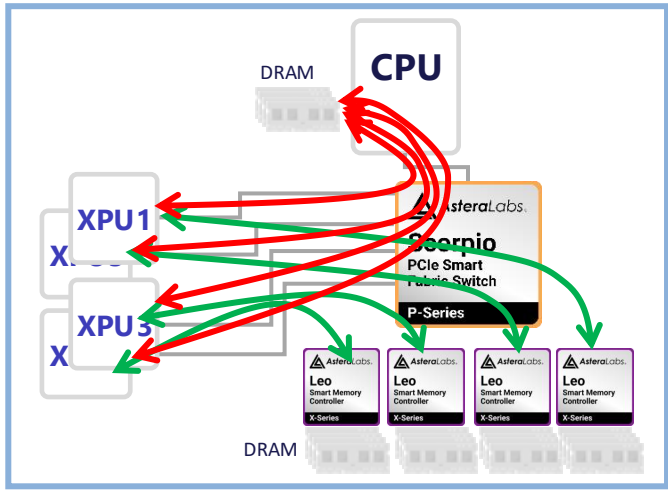
* Representative sizes only.

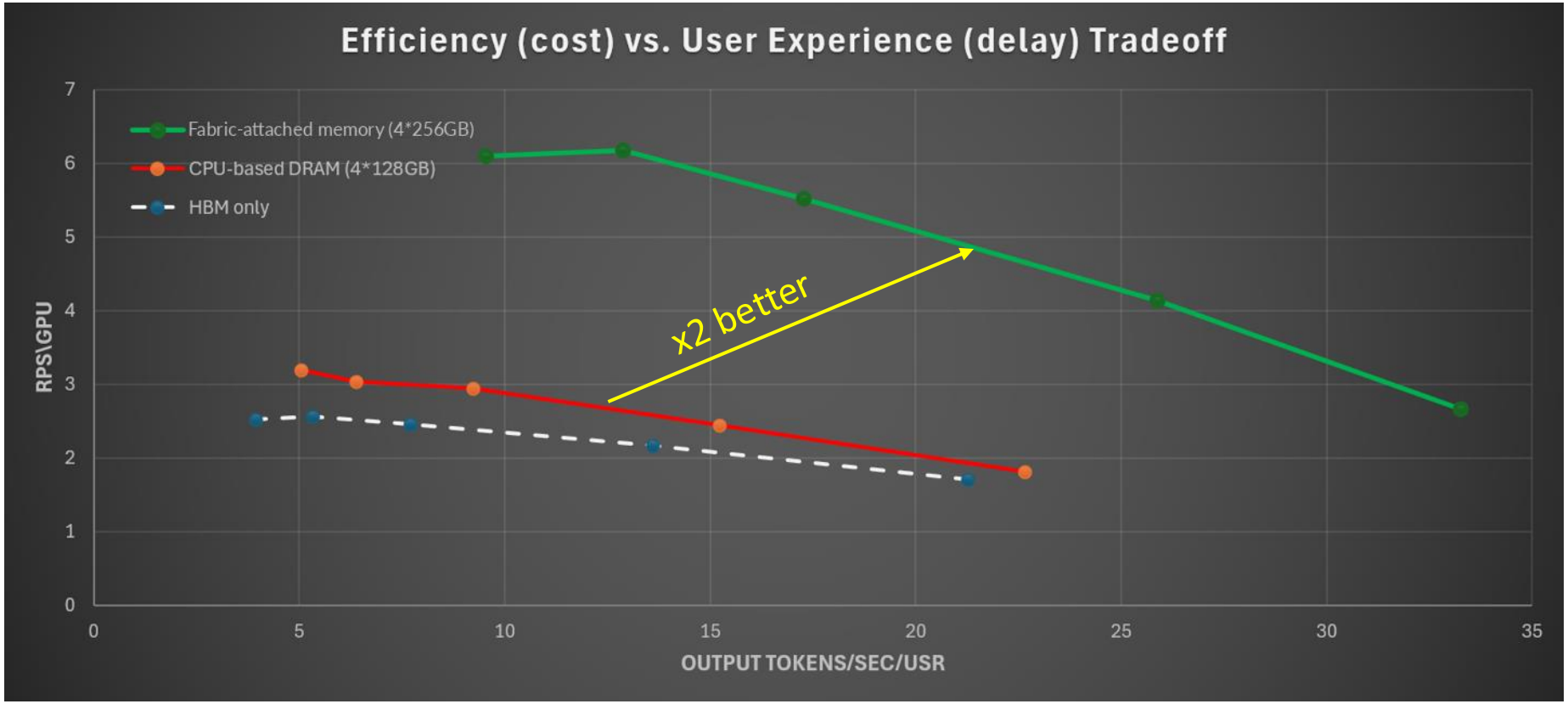
With a significant longer runtime (and cost), the system could have held 3TB on the CPU and 4TB on the memory controllers

KV-CO Results – 2

- Slightly longer prompts (2-3.5K) → additional memory is required
- Memory controllers are a convenient way to scale (for cheaper DIMMs)

Legend:
Traditional KV-CO
Fabric-attached memory

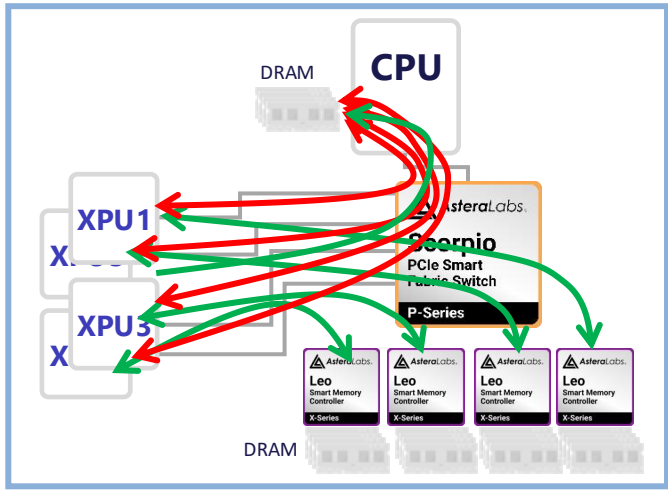




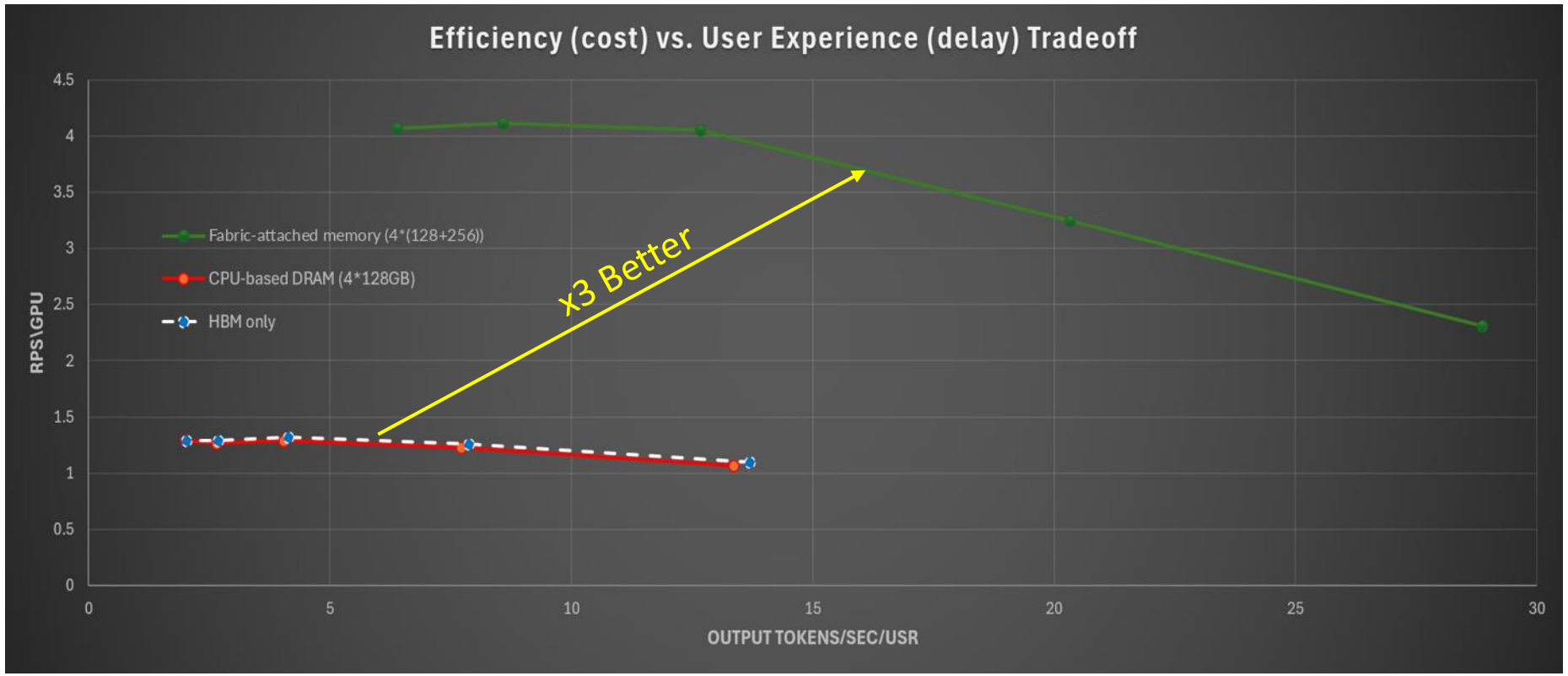
Parameters:
- Medium input prompts
- 4*128GB vs. 4*256GB

KV-CO Results – 3

Legend:
Traditional KV-CO
Fabric-attached memory



- As the Avg. prompt grows → scaling memory in the system enables KV-CO to support longer prompts
- Here, we combine memory controllers with CPU memory (to scale at all)



Parameters:
- Med-long input prompts (5-6.5K)
- 4*128GB vs. 4*(256+128)GB

Conclusions

KV-CO is sensitive to prompt size, number of turns, model, number of XPUs, etc.

The proposed P2P traffic to Scorpio Fabric Switch and Leo Memory Controllers:

- **Superior Performance**: Outperforms the CPU DRAM baseline.
- **Seamless Scalability**: Prevents single bottlenecks and scales memory controllers dynamically with XPUs.
- **Resource Optimization**: Enables downsizing to a single-socket CPU or repurposing it to run complex, heavy workloads.

Requires native memory fabric integration into KV-CO frameworks:

- **Completed:** Enhanced LMCache to support the underlying memory fabric.
- **In Progress:** Integrating native, standalone fabric support into vLLM (independent of LMCache).
- **Collaboration:** Inviting OCP CMS community to expand fabric integration into other ecosystems (e.g., SGLang, Dynamo KVCR).

Call to Action

- Join OCP CMS Workgroup, PCI-SIG, CXL, UALink and relevant fabric consortiums
  - www.opencompute.org
  - www.pcisig.com
  - www.computeexpresslink.org
  - www.ualinkconsortium.org
- Learn about vLLM, LMCache and similar KV-CO-related open-source projects
  - https://github.com/vllm-project/guidellm/blob/main/docs/guides/multiturn.md
- Software enhancements
  - LMCache
  - vLLM (when used without LMCache)
  - OCP CMS assistance with other KV-CO frameworks, such as SGLang, Dynamo KVCR etc. is welcome
- All products used in the benchmarking are available:
  Leo Smart Memory Controllers, Scorpio Smart Fabric Switches, RTX GPUs, Xeon CPU
- Meet the experts – **Astera Labs Booth B34**

Open Discussion

